primary coil produces a magnetic field (flux).This magnetic strength induces an EMF in the secondary coil. Normally, this happen only in AC circuits. Figure 2.15 shows the parts of the transformer.

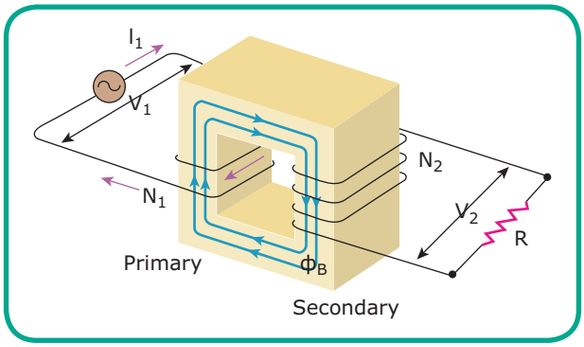


Figure 2.15 **Transformer**

The AC current flowing through the primary induces an alternating magnetic field. This magnetic field creates an EMF in the secondary. If we are giving DC in the primary, it won't induce any EMF, hence the phase does not get changed. Therefore, the transformer can be used only in AC Circuits.

2.3.2 Turns Ratio

The strength of current induced in the secondary depends upon the number of turns in the secondary coil. That is, the induced EMF is directly proportional to the turns in the coil. The voltage ratio between the primary and secondary is equal to the ratio between the number of turns in the primary and secondary.

$$\frac{E_p}{E_s} = \frac{N_p}{N_s}$$

E_p → primary coil voltage

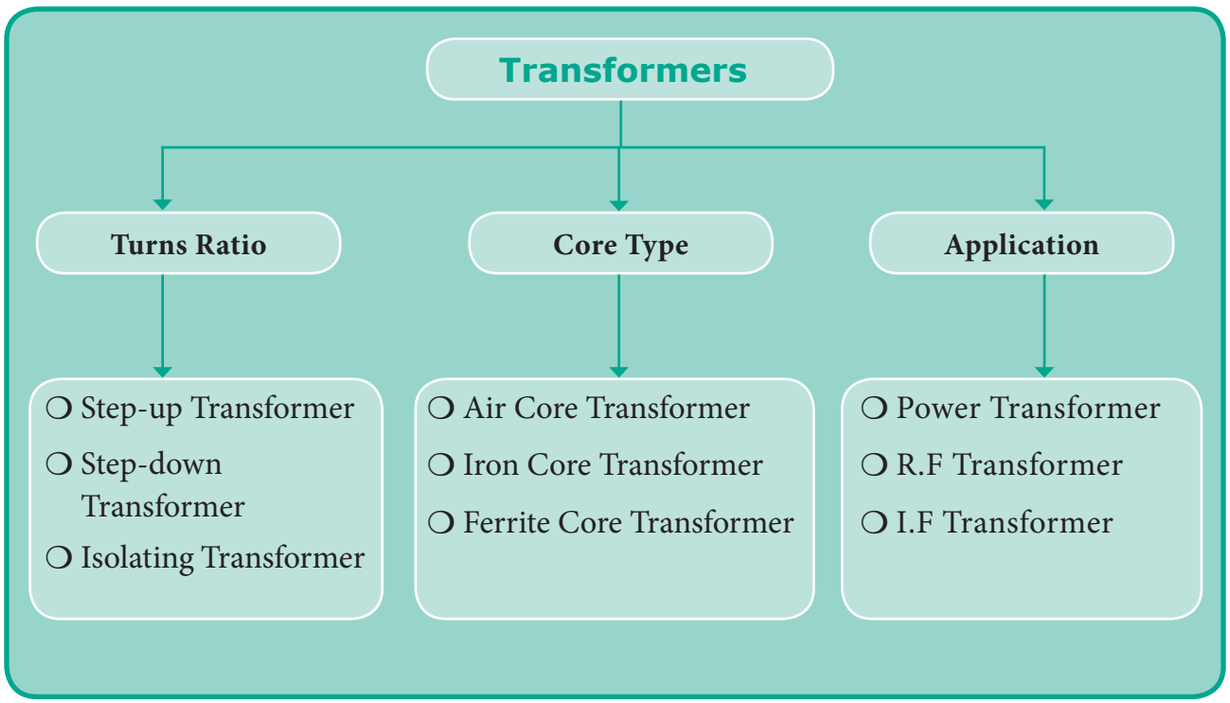
E_S → Secondary coil voltage

N_p → Number of turns in primary coil

N_S → Number of turns in secondary coil

2.3.3 Types of Transformer

Based on the turns ratio, type of the core and the type of application, the transformers are classified as given below.



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